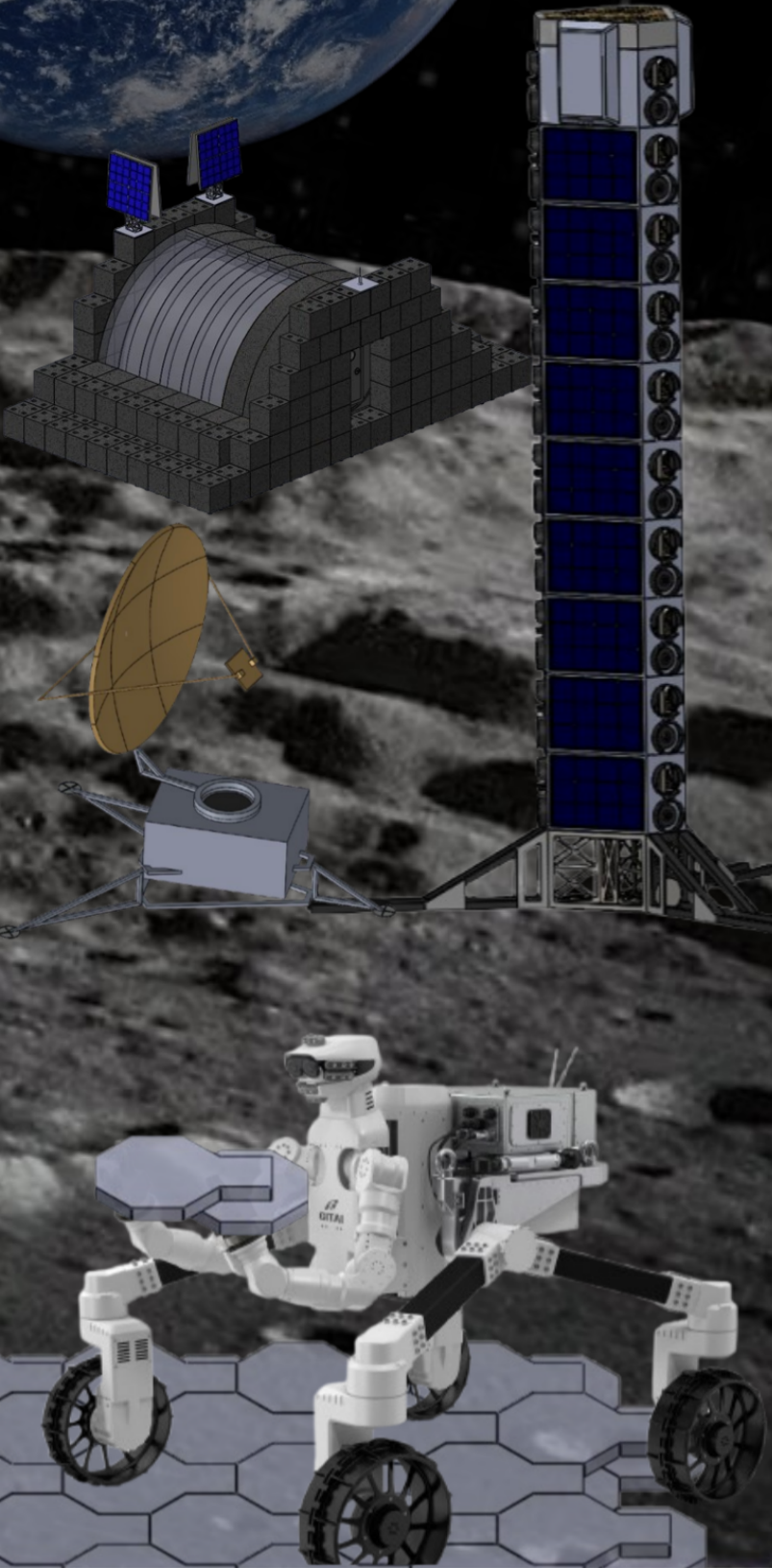


REGENESIS

Regolith Engineering for Ground Expansion,
Network Establishment, and Sustainable
Infrastructure System



2026 NASA RASC-AL Competition

Theme 4: Lunar Technology Demonstrations
Leveraging Common Infrastructure

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Contents

1	Introduction	4
2	System Architecture	4
2.1	Concept of Operations	4
2.2	Mission Timeline	5
2.3	Landing Site	5
3	Technical Approach	6
3.1	Lunar Servicing Rover	6
3.2	Communications, Navigation, Power Architecture	6
3.2.1	Sensor Selection	7
3.3	Solar Furnace	7
3.4	Human Structure	8
3.4.1	Outer Structure	8
3.4.2	Internal Pressurized Membrane (Inflatable Liner)	9
3.4.3	Validation	9
3.5	Launch Pad	9
4	Budgets	10
5	Feasibility and Risk	11
6	Conclusion	12
A	Appendix A: Supporting Calculations	13
A.1	Surface Links (S-band, 2025–2110 MHz)	13
A.2	Orbiter Backhaul (K-band, Worst-Case, Within Phased Array 60° Half-Angle)	14
A.3	Human Structure Thermal Calculations	15
B	Appendix B: Additional Figures / Tables	16



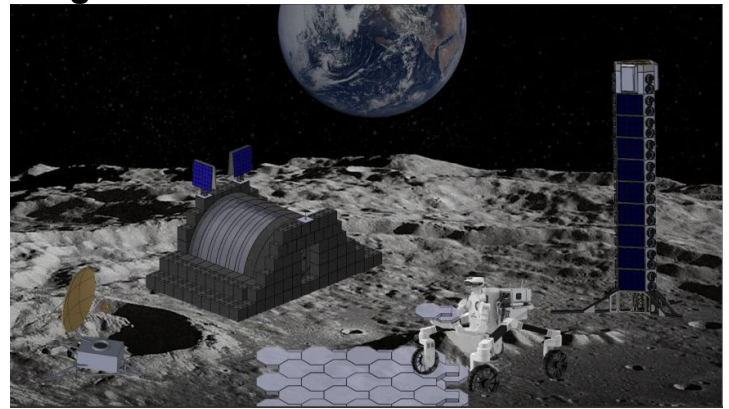
Objectives & Technical Approach:

- Leverage future LunaNet constellation to establish ground-based communications and navigation infrastructure
- Implement a solar furnace to utilize sunlight near the Shackleton Crater, In-Situ resources towards developing novel concepts
- Accomplish M2M objectives LI-4, LI-5, LI-7

Key Design Details & Innovations:

- Common Mass: 882.3 kg
- Common Volume: 3.25 m³
- Power Draw (mean): 1400W
- Power Consumption: 1400.5W
- The development of the solar furnace will maximize the capture of the sun's thermal output to produce the strongest possible building material from In-situ resources for the development of future lunar infrastructure
- The communication and navigation architecture used in this mission can be scaled for future missions and interlunar habitats

Image:



Schedule:

2027-2035	Research and Development
January 2035	Launch of Payload 1 to Lunar South Pole
January 2035	Completed Solar Furnace Construction
April 2035	Launch of Payload 2 to Lunar South Pole
April 2035	Start of Tower Construction and Launch Pad
September 2035	Estimated completion of the Launch Pad
October 2035	Launch of Payload 3 to Lunar South Pole
January 2036	Estimated completion of the Human Structure
2036-2043	Data Collection Period and Systems Validation

Cost:

\$216.4 M	\$8.9M	\$13.5-33.7M	\$259M
Launch Costs	Operating Costs	Infrastructure Costs	Total Costs

2026 RASC-AL Proposal Compliance Matrix

Lunar Technology Demonstrations Leveraging Common Infrastructure

General Proposal Information	Y/N
Does your submission adhere to the requirements and constraints of the selected topic and the design competition? Does it adhere to the page limitations and formatting guidelines?	Y
Have you proposed synergistic applications of NASA's planned current investments with new innovative approaches / capabilities / technologies to support your proposed solution?	Y
Does your proposal present original engineering analysis of innovative approaches, capabilities and/or new technologies for evolutionary architecture development to enable future missions, reduce cost, and/or improve safety?	Y
Have you provided sufficient justification for your assumptions?	Y
Have you addressed reliability and human safety considerations when trading various design options?	Y
Is your approach guided by sound systems engineering and architectural trades?	Y
Have you documented and justified your technology readiness levels (TRLs) for the technologies associated with your proposed solution?	Y
Are appropriate assessments of costs for technology maturation, system development, and production & operations included in your proposal?	Y
Did you include an appropriate project schedule and test plan with realistic development and annual operating costs (i.e., budget for technology maturation, system development, and production & operations)?	Y
Lunar Technology Demonstrations Leveraging Common Infrastructure Theme	Page #
Where does your proposal describe the Lunar Infrastructure Objectives that your concept responds to?	5,7,9
Where does your proposal describe the technology demonstration campaign your concept will enable?	3,4,7,8,9
Where does your proposal describe the support to both an early lunar ecosystem as well as pathways to industrial-scale operations and/or a robust lunar economy?	3,4,9, 11
Where does your proposal describe the infrastructure needs that would be leveraged by your campaign?	5,6,8
Where does your proposal describe the common services that would lower mass, cost, and complexity for your campaign?	3,5,6

1 Introduction

REGENESIS, Regolith Engineering for Ground Expansion, Network Establishment, and Sustainable Infrastructure Systems, is CU Boulder's solution to **Theme 4** of NASA's annual RASC-AL competition. This project aims to demonstrate the feasibility of utilizing in-situ resources to support infrastructure needs for a future lunar economy as described in NASA's Moon to Mars objectives. In this three-phase concept, solar energy is used to produce modular regolith bricks and demonstrate repeatable surface construction. The system integrates a solar furnace, autonomous rover-based assembly, and a communications/charging tower to build a launchpad and a regolith-based shelter. It is designed to prove concepts with high levels of in-situ resource utilization (ISRU), as reducing reliance on Earth-supplied materials is critical for long-term lunar operations and developing infrastructure for a future lunar economy.

2 System Architecture

2.1 Concept of Operations

There will be three payloads sent in succession, each setting up the foundation for the next payload as well as demonstrating the feasibility of lunar habitation through outlined tech demonstrations.

A lunar lander will arrive with the first payload, a rover and solar furnace that is used to create lunar regolith bricks. An RF module is also included in the first payload to establish basic communications between the solar furnace, rover, and LunaNet orbiter, validating communication for future operations. The solar furnace and communication module will be deployed by the rover that uses a ground-leveling attachment to ensure a stable foundation. From this, multiple different bricks will be produced using varying heat/cure times, and approximately forty bricks will be sent back to Earth with the first lander, as part of the solar furnace technology demonstration, for testing.

The second payload will include the rest of the communications tower to set up a permanent, self-powered contact point to establish communications and rover charging capabilities, a key technology demonstration. The tower is constructed using GITAI's inchworm robots [1]. This lays the foundation to construct a launchpad with a modular brick design, addressing a key Moon-to-Mars (M2M) infrastructure goal. The launchpad will reduce dust kick-up during descent and ascent, improving operational proximity safety and reliability. This lander will be sent back to Earth carrying multiple launchpad pavers to again test structural and thermal capabilities.

The third payload will land on the previously developed launchpad, bringing three more molds and an inflatable membrane. These molds are used to create regolith bricks, which are used in the construction of a $\frac{1}{4}$ scale human shelter concept. The regolith bricks surrounding the human infrastructure are intended to protect the habitat from radiation, thermal cycling, and micrometeoroids. An inflatable liner is inserted into the brick structure, providing a simulated earth-like environment, pressurized by pressure canisters integrated with the inflatable membrane. Further objectives include collecting long-term structural data to quantify how the structure responds to the lunar environment and assist in identifying failure modes early. Sensors measure selective ECLSS and structural parameters in and around the infrastructure and transmit them back to Earth through the tower via a monopole antenna. After the third lander leaves the lunar surface, the rover will collect multiple launchpad pavers and send analysis data back to Earth as part of the third technology demonstration.

The resulting ecosystem will be comprised of a human structure demonstrating a habitable environment, a communication architecture that supports future expansions, and a stable launchpad for continuous use and in-space mobility. This framework produces a basis for an adaptable future ecosystem that primarily relies on in-situ resources for further economic developments.

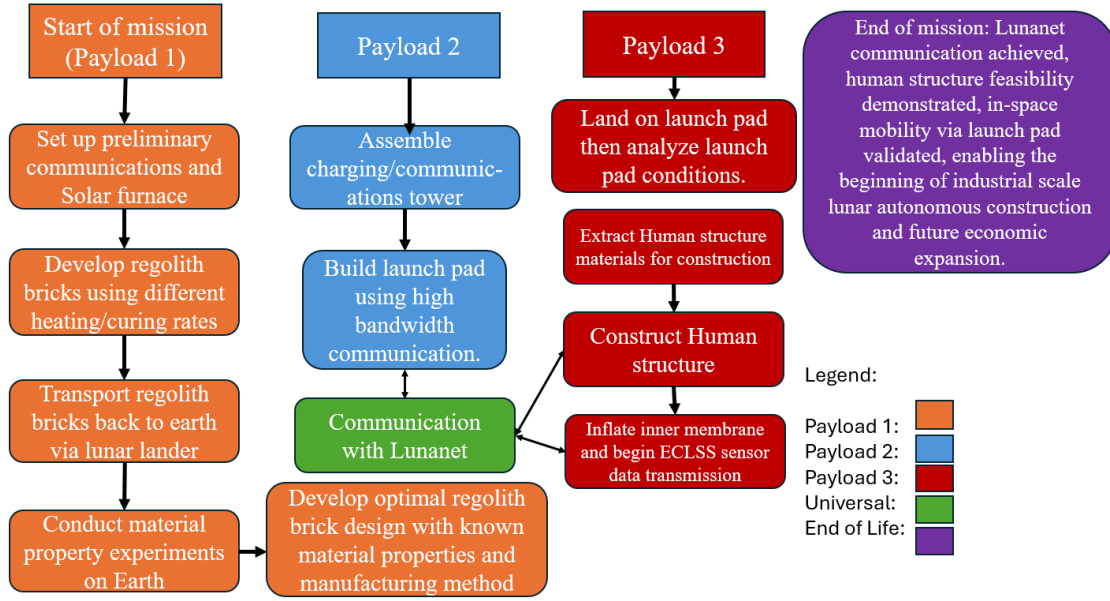


Figure 1: Full Launch Pad Assembly

2.2 Mission Timeline

Since REGENESIS leverages LunaNet for the communication architecture, the timeline is reliant on the deployment of this architecture. The constellation that comprises LunaNet is projected to be fully operational by 2030, but REGENESIS will launch in 2035 to account for unforeseen delays, R&D, and setbacks similar to those seen by the Artemis missions. SpaceX estimates that it would cost about \$100,000 [2] per kg, thus making it cheaper to launch within this timeframe. The rest of the timeline is dependent on the time required for setup, structure construction, paver creation, and the necessary window for data collection.

2.3 Landing Site

The surface operations for the REGENESIS mission are strategically situated near the rim of the Shackleton Crater at the lunar South Pole. This location was deliberately selected to align with current Artemis mission architecture, providing optimal conditions for the proposed technology demonstrations and acting as a stepping stone for future lunar operations. The primary landing site and launch pad, accommodating Payloads 1, 2, and 3, are located in an exceptionally flat region exhibiting slopes of less than 1° . This low-gradient terrain is critical for ensuring highly stable, reliable landings and ascents. Furthermore, situating the base of operations in this flat area significantly minimizes the power output required for rover traversing and

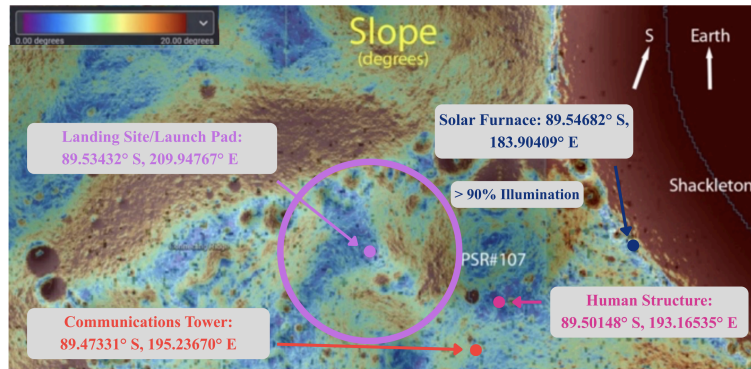


Figure 2: Map of the REGENESIS landing site and distributed infrastructure

This low-gradient terrain is critical for ensuring highly stable, reliable landings and ascents. Furthermore, situating the base of operations in this flat area significantly minimizes the power output required for rover traversing and

communication, maximizing navigational efficiency before the rover approaches more challenging topography.

To optimize the performance and safety of the mission’s distributed assets, the remaining infrastructure is placed according to specific, localized environmental advantages. The solar furnace is positioned on the edge of the Shackleton rim, approximately 5 km from the landing site, capitalizing on an area that receives over 90% solar illumination throughout the lunar year to ensure almost continuous power generation.

To support robust data relay, the communications tower is situated midway between the landing site and the solar furnace on a manageable 5° slope. Finally, the human structure is located adjacent to the communications tower on the edge of PSR #107. This location intentionally experiences significantly lower luminosity, shielding the crew and hardware from extreme solar flux. Proximity to the PSR provides a vital foothold for future volatile extraction and extended habitation goals.

3 Technical Approach

3.1 Lunar Servicing Rover

The rover selected for REGENESIS must be robust and sustain environmental stresses, such as thermal cycling and dust buildup throughout the entire timeline of the mission.

The rover must be able to lift and place objects of various sizes with precision to properly fabricate the planned structures. For these reasons, GITAI’s Rover R1.5 was selected for REGENESIS operations. The rover is capable of lifting objects of various sizes with precision sensing and placement. This rover and its inchworm IN2 arms used for building have a TRL of 7 [3] [4].

This rover can reliably conduct the tasks set forth by REGENESIS, as it’s proven to work in extreme environments. A second rover would not be needed, but it will be investigated in Phase 2 if more complex maneuvers are required or if any rover failures occur.



Figure 3: R1.5 Rover

3.2 Communications, Navigation, Power Architecture

The communication architecture for REGENESIS will be a local point of contact enabling high-speed data transfer and communication capabilities between Earth and the lunar surface. The solar panel tower designed by GITAI [5] is selected for its power output capabilities of approximately 1.2 kW with a 5 m tall tower as well as a fully modular design allowing simple implementation of a communication module. Recently demonstrating a full assembly of this tower in lunar-like conditions, this tower supports a TRL of 6 with plans to raise it to a 7 in the near future.

This architecture leverages and is aligned with the LunaNet Interoperability Specification (LNIS) and NASA Space Communication and Navigation (SCaN) standards [6] [7]. With LunaNet expecting to have a functional orbiter ecosystem by 2030, the tower is implemented with store-and-forward data handling aligned with LunaNet delay/disrupt-tolerant networking (DTN) services to support interoperability with this architecture. Overall, this system supports LI-8 through the support of autonomous rover construction and LI-7 by enabling monitoring of scientific data.

The tower-rover and tower-surface-asset communication links operate in the S-band range of 2025-2110 MHz, with surface assets (excluding the tower) utilizing singular monopole antennas with a TRL of 9. The tower itself includes 3 fixed, slanted left/right, and vertically polarized panel antennas with multiple-in multiple-out (MIMO) functionality to support expansion, boost data speeds, and increase reliability. These are spaced 120° to provide 360° signal coverage. The 5 m tower height

provides a line-of-sight range of approximately 6 km with the rover antenna at 1 m. This provides a range larger than the intended construction area, allowing for future expansion. With the max expected distance between the rover and tower being 4 km, the calculated energy per bit to noise power spectral density ratio indicates significant overhead and the capacity to increase data rates if needed.

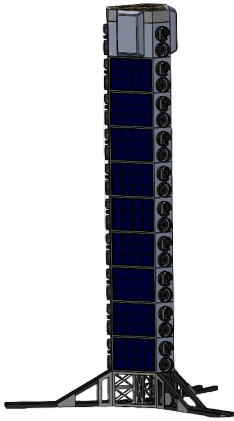


Figure 4: Communications and Power Generation Tower

High data throughput is achieved by utilizing K-band forward and return allocations, filtered OQPSK modulation, and LDPC coding, all defined in LNIS v5 (PFK3-O and PRK3-O) [6]. Supporting multi-Mbps tx and rx capacity [A.2], this architecture provides a scalable communication system that supports early demonstrations and future expansions. For mission safety, the rover is also equipped with a low-power, 8 dBi gain, S-band patch antenna mounted on the top, in the 2025 – 2110 MHz forward and 2200 – 2290 MHz return allocations defined in LNIS v5 [6]. This supports emergency low-rate signal uplink and downlink using frequency-division duplexing (FDD) in the event of tower malfunction or terrain obstruction. Under nominal conditions, data is routed to the orbiter through the tower.

Surface operation requires approximately 12 W [A.1] of power. The panel antennas utilize RF switching to avoid multiple parallel transmit chains that would draw additional power and add complexity. The K-band backhaul link using a phased array requires approximately 30 W of electrical power, primarily due to beamforming, the power amplifier, and expected inefficiencies. During simultaneous surface and orbiter operations, total communications power consumption is expected to remain below 50 W.

For power storage, lithium-ion batteries were selected due to their high specific energy, efficiency, and TRL of 9 [8]. The tower incorporates two energy storage sections. The first consists of a 6 kWh charging bank (2 x 3 kWh battery), and the other is a 2 kWh communication reserve (2 x 1 kWh battery). Assuming the rover has 2.8 kWh of usable energy from a 4 kWh onboard battery and that the charger is capable of delivering 400 W at 85% transfer efficiency [9], the rover will take approximately 8.5 hours to charge while providing between 3.5 to 14 hours of life depending on the average power consumption. To survive the lunar night, the batteries will be placed into a pack and experience low-power resistance heating to maintain a livable temperature. A Minko HAP6731 resistance heater, with an approximate TRL 8, will be used to fulfill the heating requirements for the batteries. The current charging module that will be used is the Astrobotic Proximity Charger with roughly TRL 6 after a recent flight test [10]; however, this mechanism will be expanded upon and accounted for in the next phase [11].

The system is not sized for full operational capacity through extended darkness. Instead, it transitions to a survival mode, maintaining critical communication and thermal survival power loads. After a 48-hour period without solar input, the system transitions from reduced power to hibernation, solely emitting a health status update periodically to the orbiter.

3.2.1 Sensor Selection

3.3 Solar Furnace

ISRU is a fundamental requirement for sustainable lunar surface development. In this project, lunar regolith is the primary construction material. Unprocessed regolith is not suitable for structural use, but via thermal processing, suitable strength can be achieved for a variety of applications. The proposed solar furnace solves this problem.

There are three main components for the solar furnace, including (1) the concentrator, (2) the furnace body, and (3) the electrical system. The concentrator is made up of two separate mirror arrays. The primary

Sensor Name	Price	Weight (g)	Rationale
Dust Sensor (Nova SDS011)	\$20 - \$30	50 - 70	Uses laser scattering to count particles, > .3 um, to detect air contamination.
Pressure Sensor (Bosch BMP280)	\$10	1.3	Barometric sensor that monitors the habitat's internal pressure to ensure a stable atmosphere.
Pavers Sensor (HBM M-Series)	\$25	Negligible	NiCr foil strain gauges rated for -20°C - +300°C to monitor mechanical strain in sintered regolith.
Thermal Radiation (Omega OMEGACLAD XL)	\$135	453.6	MgO-insulated Type K/N thermocouple with Inconel sheathing, used to detect thermal radiation.
High Energy and Proton Sensor (ESA SREM)	\$200	2500	MgO-insulated Type K/N thermocouple with Inconel sheathing, used to detect thermal radiation.

mirror array is attached to a rotating plate that tracks the sun and sees $3m^2$ of incident sunlight. It is divided into nine sections to reduce transport volume and will be assembled on the lunar surface. The secondary array reflects the output onto the "receiver puck," an element of the solar furnace. The product of the combined mirror array is shaped such that it concentrates the sunlight into a near-uniform strength across a circular area.

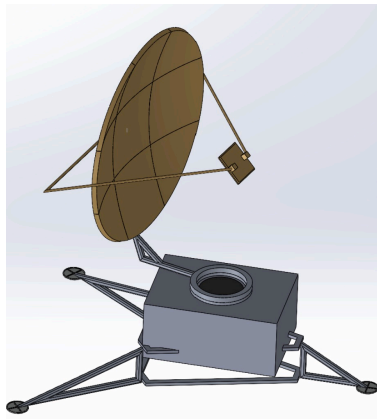


Figure 5: Solar Furnace

The mirror surface will be constructed from advanced micro sheet glass for durability[12]. This protects panels during transport, and the multi-panel design is constructed from 3 unique panels. The concentrator arm has a hinge mechanism, allowing the arm to enter a lowered position, where the rover can service the panels. This position is used during assembly and also allows for servicing. Servicing includes regular cleaning to avoid dust accumulation losses and also allows for swapping out damaged panels.

The furnace body facilitates the various heat transfer processes required for operation. First, the concentrated sunlight is absorbed by the receiver puck and spread to the mold's upper surface by a thin layer of diamond. The 2-layer mold then conducts heat to the regolith. The outer layer of the furnace body is thermally isolated from the receiver puck and mold to reduce losses during heating.

The mold consists of an inner layer made from yttrium-stabilized zirconium (YSZ) and an outer layer made from alumina. The YSZ contacts the regolith directly, as it can handle the corrosive properties of regolith, while the alumina outer layer provides good durability for the thermal cycles seen during the melting process.

The electrical systems required to drive the solar furnace are low-complexity when compared to similar sintering or microwave methods. Solar panels are mounted to the primary array to supply power. A hollow central bearing is connected to the main drive motor, and a larger outer bearing provides support to the concentrator arm. The solar array feeds power to the computer module at the base of the furnace. The computer module is buried in a well-insulated shell, that provides a stable temperature for the battery and electronics.

10 heating profiles will be tested during the payload one mission. Heat rate will vary from 2-6°C per minute, temperature between 1100-1200°C, and cooling rate between 5 and 10°C [13].

3.4 Human Structure

3.4.1 Outer Structure

The brick structure technical demonstration aims to demonstrate that sintered lunar regolith bricks can build robust structures, specifically for lunar housing. This structure aims to demonstrate that lunar rovers can build complex structures that could support a future human presence and economy on the lunar surface (LI-4). This structure also aims to demonstrate the pavers' ability to produce these lunar housings quickly and efficiently, showcasing "industrial ISRU capabilities" (LI-7).

These brick structures are versatile housings intended to protect the inside from the harsh thermal and radiation conditions found on the lunar surface [14]. The structures have a wide range of applications, such as housing for astronauts, manufacturing facilities, roads, or launch pads. These structures utilize a 3-brick design to minimize the weight and cost of the sintering materials and molds, as well as to provide simplicity and versatility to the structures.

This design uses a cubic brick that makes up the walls of the structure, an arch brick that makes up the ceiling, and a connector block that connects the arches to the cubes, which is all held together by several interlocking mechanisms. The bricks weigh about 16.6 kg each and are 5035 cm^3 .

3.4.2 Internal Pressurized Membrane (Inflatable Liner)

The inflatable liner provides a pressurized, habitable volume within the brick structure. For deployment, a regulated compressed-gas tank is used to inflate the membrane atop the base layer of bricks after anchoring to the base connection interface[15]. The membrane consists of four key layers: a crew-safe inner liner, a gas bladder to retain the pressurized atmosphere, a restraint layer to help carry internal pressure loads, followed by a SMPC outer rigidization layer to stiffen the inflated habitat after deployment [16]. Ingress/egress will be provided by an NDS-class, circular hatch ring integrated at the end cap.

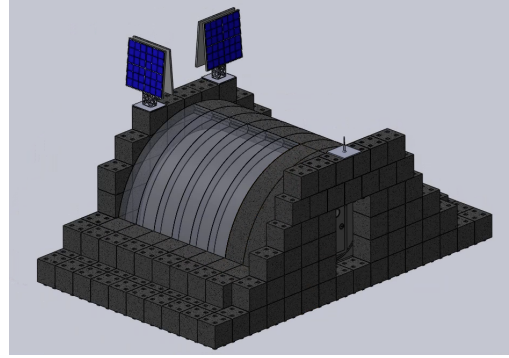


Figure 6: Depiction of the Human Structure

3.4.3 Validation

There are two main factors to validate the structure's effectiveness. The first is structural stability. This structure needs to be able to withstand the stress from its own weight. Additionally, this structure will need to withstand collisions from micrometeoroids, as well as moonquakes. This will be analyzed by using strain gauges and visual inspection from the rover's LIDAR to detect significant displacements or fractures. The second is thermal and radiative insulation. The major purpose of this structure is to reduce the insulation load placed on the internal membrane. Pressure, dust, thermal, and high-energy radiation sensors as described in Table 1 will line the inside of the inflatable structure. They will draw power from solar panels located outside the structure via magnetic connections. They will then transmit data to a monopole antenna, which communicates directly with the tower. The inductive connections will be explored in Phase 2.

Preliminary structural simulations were conducted in ANSYS to determine the maximum tensile and compressive stresses due to structure weight and internal membrane pressure. The maximum tensile and compressive stresses are 2.8 MPa and 1.5 MPa, respectively. These results indicate that the structure would be stable under steady and transient stress conditions. Heat transfer analysis yielded that radiation largely does not transmit through the inflatable layer. Should this project advance to Phase 2, a formal thermal simulation will be conducted, and if radiation is found to leak, Kevlar bagging will be investigated to prevent it.

3.5 Launch Pad

Although modern landers have proven land and launch capabilities on the lunar surface, the rocket fumes and force generated by the boosters cause soil erosion of up to 100 meters deep and dust kick-up [17]. A launch pad solves for this by blocking lander fumes from interacting directly with the lunar surface. The regolith

bricks that will be produced throughout the duration of this technology demonstration can be successfully used as the primary construction material of lunar launch pads. Their high compressive strength and low thermal conductivity make them the ideal ISRU material. This structure aims to demonstrate that the sintered bricks can comprise a launch pad geometry that can be scaled to support the launching and landing of industrial-scale lunar landing vehicles (LI-5).

Structural performance, flow penetration, and heat propagation indicate launchpad success. Ultimately, the paver chosen for REGENESIS is the one shown in figure 7, which uses only one unique paver shape, eliminating the need for multiple molds for the construction of the lunar launchpad. The interlocking scheme presented below greatly prevents flow penetration. Thermal propagation has been studied extensively via binding agents with thermal conductivities of around 0.29 [18]. Rocket motors have been tested on these composites and demonstrated minimal thermal propagation and defects, achieving a TRL of 5 for this technology [19]. Instead, REGENESIS will use pure regolith blocks to reduce additional Earth-based construction dependence. The baseline thermal conductivity of a pure regolith block is about 0.26 [20], which is similar to the RTV composite. Given the similar thermal conductivities, the pure regolith block is expected to provide close thermal propagation results while eliminating binding agent dependence. The thickness of this design was increased to account for higher temperatures due to larger, more advanced landers.

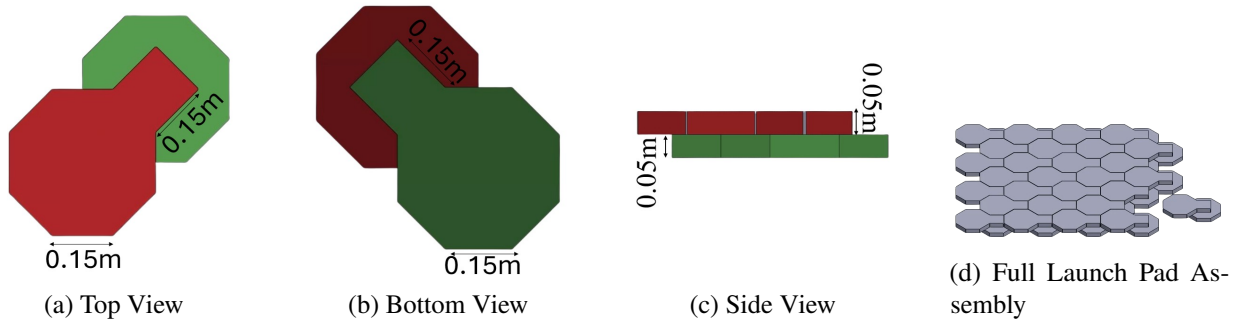


Figure 7: Launch Pad Paver Design

The force output from a Blue Origin MK1 was used to analyze the structural performance. The max deformation from the lander legs was 3.59×10^{-7} m and from the thruster was 1.63×10^{-7} m. If this project advances to the second round, a proper CFD simulation and thermal simulation will be used to formally determine flow penetration and heat propagation. Additionally, stability of the pavers was assumed to be achieved from the interlocking scheme, preventing unwanted movements. However, if it's determined that the launch pad is unstable, REGENESIS will investigate stability through sintering in the next round.

4 Budgets

The budget of REGENESIS is subdivided into three payloads. Power estimates in Payload 1 represent power drawn, while Payload 2 power estimates represent power generated. The expanded cost of the systems can be shown in Table [3] with margins provided. The margins are based on confidence levels represented as: low: 70%, medium: 40%, high: 20%, High TRL data sheet reference: 0%. The majority of mission costs will be from lander estimates, which are about \$1.2 million per kg today. However, when REGENESIS launches its first payload in 2035, it is expected that launch costs will lessen to \$100M per metric ton or \$100,000 per kg as listed by SpaceX [2]. Table [3] costs have been adjusted for inflation at a rate of 2.5% (average inflation rate over the past 20 years). A few assumptions have been made when deriving estimates. The yttria-stabilized zirconia molds are material costs with a 10% added margin for labor and R&D. Material property testing costs depend on the tests taken, the research company, and the accuracy of the tests. For

inexact or unavailable cost estimation, the Parametric Cost Modelling Process was implemented to achieve rough cost estimates. The leading example of this is the GITAI rover, which was estimated via comparison to older rovers of similar capability, such as Astrobotics' CubeRover. For the operating costs, Mars Exploration Rovers public data was used. A total of \$335.8m in operational costs over the course of 15 years [21] leads to an estimate of \$22.8m in yearly expenditures for a Mars mission of similar design. Several factors decrease this value even further, such as planning, infrastructure, communication, and operational staffing costs. The difference in requirements and required labor leads to a decreased percent estimate of approximately 60% or a final value of $\approx 8.95\text{m}$.

Component	Mass (kg)	Margin (%)	Vol. (m^3)	Margin (%)	Power (W)	Margin (%)	Cost (\$)	Margin (%)
Payload 1								
GITAI Lunar Rover (1)	280	0 [3]	1.6866	0 [3]	115–1500	0 [3]	6,354,830.87	40
GITAI Inchworm IN2 (2)	50	0 [4]	0.2516	0 [4]	60–200	0 [4]	1,270,966.17	40
Charging Tower RF Module (1)	57.23	40	0.03	40	126	0 [22]	1,477,434.17	40
Solar Furnace (1)	158	70	0.29	70	327.12	70	478,000	70
YSZ Launch pad Mold (1)	36	20	0.04	20	–	–	21,352.23	20
Offloading Ramp (1)	92	20	0.4	20	–	–	69,903.14	20
Earth Regolith Brick Testing (10)	–	–	–	–	–	–	10,000	40
Payload 2								
Charging Tower Module (8)	44.63	40	0.02	40	133–144	20	1,270,966.17	40
Charging Tower Base (1)	170	40	0.07	40	–	–	1,906,449.26	40
Leveling Roller (1)	71	20	0.172	20	–	–	130,000	70
Payload 3								
Inflatable Structure (1) (incl. sensors)	270	20	0.035	20	–	–	254,193.23	70
YSZ Cube Mold (1)	13.67	20	0.002822	20	–	–	8,107.92	20
YSZ Curved Mold (1)	74.116	20	0.017	20	–	–	20,979.46	20
YSZ Connector Mold (1)	16.535	20	0.018	20	–	–	16,989.02	20
Sierra Space 400 in^3 COPV (2) (incl. gas)	4.5	20	0.00655	0 [15]	–	–	38,129	40

Table 1: Payload mass, volume, power, and cost estimates with assigned margin percentages based on design maturity.

Payload	Mass Low	Mass High	Volume Low	Volume High	Power Range (W)	Cost Low (\$)	Cost High (\$)
Payload 1	564.138	882.322	2.6466	3.2526	360–2079	6,057,692.48	16,029,213.05
Payload 2	373.024	823.056	0.2756	0.5284	745–1498	7,283,507.19	17,124,850.12
Payload 3	306.6568	459.985	0.071357	0.1004864	–	158,873.86	594,181.32

Table 2: Payload totals for mass, volume, power, and cost. Bounds reflect uncertainty margins applied at the subsystem level.

Common mass and volume envelope: 882.322 kg, 3.2526 m^3 .

Total Costs (launch & payload): Lower Estimate: \$251,196,526.46; Upper Estimate: \$298,444,844.50.

5 Feasibility and Risk

Figure 8 presents REGENESIS systems risks. The critical risks associated include power infrastructure, communications, and rover operation failures. During the lunar night, battery performance and lifespan are ad-

versely affected as temperatures fall below the allowable operating range. This effect can be mitigated through the use of a low-power resistive heater. The resistive heater would preserve battery degradation and greatly reduce the chances of a power infrastructure failure. There's excess mass in payload 3 to support contingency plans and redundancies should any infrastructure be damaged on the lunar surface. A communication failure would cut all data from being transmitted to and from the lunar surface. Such an event is unlikely since communications modules are stationary in the tower and could reliably send data to LunaNet. Rover-related systems are a major point of failure for the technology demonstrations. With a TRL 7 maturity rating, it is assumed that a rover meant for lunar operations would handle thermal cycling, dust, and radiation as part of its mechanisms.

The tasks assigned to the rover are simple and well within a margin of safety. Additionally, the coordinates chosen for REGENESIS ensure the rover doesn't traverse extreme gradients where it could experience extreme power loss or mechanical failure. Lastly, faulty landing conditions can be catastrophic for the mission, and damage payload components.

This risk is being addressed by first having the first two payloads land on a relatively flat area, and for the third to land on a relatively stable launch pad.

Consequence	5	8,4,10,	5	9			1. Micrometeoroid impacts
	4	1					2. Dust accumulation
	3	3,7					3. Structural Failure
	2	11	2				4. Battery Degradation
	1		6				5. Rover Breakdown
		1	2	3	4	5	6. Sensor Failure
		Probability					7. Pressurization Failure
							8. Power Failure
							9. Non ideal landing conditions
							10. Communications failure
							11. Line of Sight loss

Figure 8: Risk Matrix

6 Conclusion

REGENESIS's technology demonstrations validate a scalable framework for sustained lunar progress and define requirements for the next advancement in lunar colonization. The "test bricks" produced by the solar furnace will create a comprehensive dataset of material properties depending on casting parameters, leading to the development of reliable and highly accurate material models that support larger-scale, industrial, in-situ construction. The communications tower and LunaNet interoperability establish an adaptable communications and power baseline, where additional towers can be incrementally added to increase generated power, communication bandwidth, and redundancy, directly addressing critical failure points. The launch pad technology demonstration will provide safe, repeatable transport to and from the Moon, marking a turning point in logistical capacity and launch frequency. The LIDAR analysis conducted by the rover on the launch pad pavers helps define safe operational mass limits for lunar landers. Higher payload capacities can be developed in reference to these bounds, thus maximizing payload capacity while preserving robust factors of safety for in-space transport systems. Finally, the lunar habitat was implemented to determine whether regolith structures in tandem with terrestrial imports could sustain temperature variation, radiation, and minor structural anomalies. The integrated sensors within the inflatable membrane will record crucial ECLSS data, which can be used to define top-level requirements for future lunar habitat developments, paving the way for industrial-scale ECLSS projects. REGENESIS does not merely demonstrate feasibility; it delivers innovative strategies while generating the operational data needed to enable the next era of lunar expansion.

A Appendix A: Supporting Calculations

Communications Link Budget Summary

Link budgets were evaluated for S-band surface communications and K-band orbiter backhaul. Orbiter waveform and allocations follow LNIS v5 Proximity K-band interfaces (PFK3-O / PRK3-O).

A.1 Surface Links (S-band, 2025–2110 MHz)

Assumptions

- Frequency: $f = 2067.5$ MHz
- Tower antenna gain: $G_t = 12$ dBi (120° sector)
- Surface asset gain: $G_r = 2$ dBi (monopole)
- Lumped losses (implementation + orientation mismatch): $L = 3$ dB
- Tower transmit power (approximation): 2 W (33 dBm)
- Surface transmit power: 1 W (30 dBm)

Received power is computed as:

$$P_r = P_t + G_t + G_r - L - L_{\text{path}} \quad (1)$$

Tower–Rover (4 km): $L_{\text{surface}} \approx 120$ dB

$$P_r = -76 \text{ dBm (tower} \rightarrow \text{rover)} \quad (2)$$

$$P_r = -79 \text{ dBm (rover} \rightarrow \text{tower)} \quad (3)$$

Tower–Human Infrastructure (1.02 km): $L_{\text{surface}} \approx 105$ dB

$$P_r = -60 \text{ dBm (tower} \rightarrow \text{infrastructure)} \quad (4)$$

$$P_r = -64 \text{ dBm (infrastructure} \rightarrow \text{tower)} \quad (5)$$

Surface Link Margin (E_b/N_0)

$$R_b = 2.048 \times 10^6 \text{ bps} \quad (6)$$

$$10 \log_{10}(R_b) = 63.11 \text{ dB} \quad (7)$$

$$T_{\text{tower}} = 240 \text{ K}, \quad T_{\text{rover}} = 330 \text{ K} \quad (8)$$

$$\left(\frac{E_b}{N_0} \right)_{\text{dB}} = P_r^{(\text{dBW})} + 228.6 - 10 \log_{10}(T_{\text{sys}}) - 10 \log_{10}(R_b) \quad (9)$$

$$E_b/N_0 = 34.30 \text{ dB} \quad (10)$$

$$E_b/N_0 = 32.69 \text{ dB} \quad (11)$$

$$E_b/N_0 = 49.30 \text{ dB} \quad (12)$$

$$E_b/N_0 = 47.69 \text{ dB} \quad (13)$$

A.2 Orbiter Backhaul (K-band, Worst-Case, Within Phased Array 60° Half-Angle)

Assumptions

- Max range: $R \approx 186 \text{ km}$
- Frequencies: Forward 23.35 GHz Return 27.25 GHz
- Gains: $G_t = 25 \text{ dBi}$ (tower, LHCP), $G_r = 25 \text{ dBi}$ (orbiter, LHCP)
- Losses: $L = 5 \text{ dB}$ (2 dB implementation + 3 dB scan/pointing)
- Transmit power: $P_t = 2 \text{ W}$ (33 dBm)
- Noise temps (receiver): $T_{\text{sys,orb}} = 130 \text{ K}$ (fwd), $T_{\text{sys,tower}} = 280 \text{ K}$ (ret)

$$FSPL = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right) \quad (14)$$

$$FSPL_f = 165.2 \text{ dB} \quad (15)$$

$$FSPL_r = 166.5 \text{ dB} \quad (16)$$

$$P_r = P_t + G_t + G_r - L - FSPL \quad (17)$$

$$P_r = -87.2 \text{ dBm (tower} \rightarrow \text{orbiter)} \quad (18)$$

$$P_r = -84.5 \text{ dBm (orbiter} \rightarrow \text{tower)} \quad (19)$$

Backhaul Throughput

$$R_{\text{info}} = 2R_s r \quad (20)$$

$$R_s = 2.048 \text{ Msps} \quad (21)$$

$$R_{\text{info}} = 2.048 \text{ Mbps} \quad (22)$$

Backhaul Link Margin (E_b/N_0)

$$\left(\frac{E_b}{N_0} \right)_{dB} = P_r^{(dBW)} + 228.6 - 10 \log_{10}(T_{\text{sys}}) - 10 \log_{10}(R_b) \quad (23)$$

$$E_b/N_0 = 27.15 \text{ dB} \quad (24)$$

$$E_b/N_0 = 26.52 \text{ dB} \quad (25)$$

A.3 Human Structure Thermal Calculations

$$S = 1363 \text{ W/m}^2 \quad (26)$$

$$\alpha = 0.92 \quad (27)$$

$$\varepsilon = 0.96 \quad (28)$$

$$k = 0.6 \text{ W/(m} \cdot \text{K)} \quad (29)$$

$$t = 0.125 \text{ m} \quad (30)$$

Absorbed Solar Heat

$$\dot{Q}_{\text{abs}} = \alpha AS \quad (31)$$

$$A = 0.103 \text{ m}^2 \quad (32)$$

$$\dot{Q}_{\text{abs}} = 0.92 \times 1363 \times 0.103 \quad (33)$$

$$= 129.16 \text{ W} \quad (34)$$

Geometric Area Calculation

$$L = \frac{240}{360}(2\pi R) = 824.67 \text{ mm} \quad (35)$$

$$A = L \times t \quad (36)$$

$$= (824.67)(125) \text{ mm}^2 \quad (37)$$

$$= 1.03 \times 10^5 \text{ mm}^2 \quad (38)$$

$$= 0.103 \text{ m}^2 \quad (39)$$

Conduction Temperature Difference

$$\dot{Q} = \frac{kA}{t}\Delta T \quad (40)$$

$$\Delta T = \frac{\dot{Q}t}{kA} \quad (41)$$

$$= \frac{129.16 \times 0.125}{0.6 \times 0.103} \quad (42)$$

$$= 261.24 \text{ K} \quad (43)$$

$$T_2 - T_1 = 261.24 \text{ K} \quad (44)$$

Radiative Equilibrium

$$\dot{Q}_{\text{emit}} = \varepsilon \sigma AT^4 \quad (45)$$

$$\dot{Q}_{\text{emit}} = \dot{Q}_{\text{abs}} \quad (46)$$

$$129.16 = 0.96 \times (5.67 \times 10^{-8}) \times (0.103) \times T_1^4 \quad (47)$$

$$T_1 = 389.59 \text{ K} \quad (48)$$

$$T_2 = 128.35 \text{ K} \quad (49)$$

Assume a closed system: $\dot{Q}$ is conserved at $129.16 \frac{\text{W}}{\text{m}^2}$

$$A = .103 \text{ m}^2 \quad (50)$$

$$k = 0.164 \text{ W}/(\text{m} \cdot \text{K}) \quad (51)$$

$$t = 0.02 \text{ m} \quad (52)$$

$$\Delta T = \frac{129.16(.02)}{.164(.103)} = 152.92 \text{ K} \quad (53)$$

Assumption

Since $\Delta T \gg T_2$, it is assumed that the solar flux has no effect past the first rigidization layer.

B Appendix B: Additional Figures / Tables

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